MIT Academy of Engineering (An Autonomous Institute Affiliated to Savitribai Phule Pune University)	COURSE SYLLABUS	
SCHOOL OF COMPUTER ENGINEERING	**W.E.F**	**AY:** 2023 - 2024 (Rev. 2023 (NEP))
SECOND YEAR BACHELOR OF TECHNOLOGY	**COURSE NAME**	Essentials of Data Science
	COURSE CODE	2304102T
	COURSE CREDITS	2
**RELEASE DATE : ** 01/08/2023	**REVISION NO.**	1.0

TEACHING SCHEME (HOURS/WEEK)		**EXAMINATION SCHEME AND MARKS**					
		THEORY			LABORATORY		TOTAL
LECTURE	PRACTICAL	IA	MSE	ESE	CA	PRACT/DEMO/PRES.	
2	NIL	15	20	40	NIL	NIL	75

PRE-REQUISITE :

COURSE OBJECTIVES :

2304102T.CEO.1: To get familiar with the basics of Python programming.
2304102T.CEO.2: To learn different data structures in Data Science.
2304102T.CEO.3: To use data computation methods in Data Science.
2304102T.CEO.4: To introduce data manipulation methods in Data Science.
2304102T.CEO.5: To use data visualization methods in Data Science.

COURSE OUTCOMES :

The students after completion of the course will be able to,

2304102T.CO.1: Build a python program for handling syntax and semantics for any given problem[L3].
2304102T.CO.2: Demonstrate proficiency in handling data structures useful in Data Science[L3].
2304102T.CO.3: Apply the suitable methods of data computations on real time data[L3].
2304102T.CO.4: Interpret the different methods of data manipulation on real time data[L3].
2304102T.CO.5: Apply data visualization for real time data to get better insights[L3].

COURSE ABSTRACT

Essentials of Data Science (EDS) is the First Year Semester-II course. This course aims at equipping learners to be able to provide the essence of Data Science by using python programming. In this course, firstly, the learners will learn python fundamentals required for Data Science. The learners are able to use different data organizational structures to store the real time data. Data computations are required to maintain the quality of data. EDS helps to learn different data computations. Data manipulations help in handling missing and noisy values in Data Processing. The learner will learn data storing, loading, cleaning, preparation, wrangling, transformation etc. Data visualization is the practice of translating information into a visual context, such as a map or graph which tends to identify the patterns, trends from the data. This course will increase the learner's interest in Data Science. The activity based learning is adapted to learn this course.

THEORY COURSE CONTENT

UNIT 1	Python Fundamentals for Data Science	6 HOURS

App/System/Case study: Weather Analysis
Contents: Introduction, Data Types- Mutable and immutable types, Data Conditioning using Decision Statements and Iterative Statements.
Functions: Basics of functions, variable scope and lifetime, Lambda or anonymous function.
Modules: Introduction to modules, packages and standard library modules.
Data Handling using Files.
Self Study:CSV Reader and Writer
Further Reading: Excel Reader and Writer

UNIT 2	Data Organizational Structures	6 HOURS

App/System/Case study: Retail-Industry/Twitter Reviews
Contents: Categories of Data- Unstructured, Structured, Categorical and Time Series
List- Creating List, List operations
Tuple- List of Tuples, Immutability,
Dictionaries- Creating Dictionary, Adding to Dictionary with SetDefault, Loading JSON to Dictionary
Set- Removing Data from Sequences, Performing common data set operation
Self Study:JSON File Handling
Further Reading:XML File Handling

UNIT 3	Data Computation	6 HOURS

App/System/Case study: Stock Market
Contents: Data Operations- Arithmetic and Statistical, Bitwise Operators, Linear Algebra, Copying and viewing data in arrays, Stacking, Data Sorting, Data Searching and Indexing, Data counting, Mathematical Operations, Broadcasting, Matrix Operations, Structured Data.
Self Study: Fancy Indexing
Further Reading: Parallel Processing using Arrays

UNIT 4	Data Manipulation	6 HOURS

App/System/Case study: Salary Analysis
Contents: Data Loading, Data Storage, Summarizing and Computing Descriptive Statistics, Data Cleaning, Data Preparation, And Data Wrangling: Join, Combine, and Reshape, Data Transformation, Data Aggregation and Group Operations.
Self Study: Working on XML and JSON
Further Reading: Data Profiling

UNIT 5	Data Distribution and Visualization	6 HOURS

App/System/Case study: Sales
Contents: Population and Samples
Data Distribution- Statistical Analysis of Data, Data Analytics Techniques; Descriptive, Predictive and Prescriptive, Normal Distribution Significance of Data Visualization in Data Science, Plots- Line Chart, Bar Plot, Histogram, Scatter Plot, Pie Chart, Density Plot, Facet Grids for categorical data, Group Plots
Self Study: 3D Visualizations
Further Reading: Supervised Learning - Classification-KNN, Regression-Simple Linear Regression

TEXT BOOK

1. VanderPlas, J. (2016). Python data science handbook: Essential tools for working with data. " O'Reilly Media, Inc.".
2. McKinney, W. (2012). Python for data analysis: Data wrangling with Pandas, NumPy, and IPython. " O'Reilly Media, Inc.".
3. McKinney, W. (2017). Python for data analysis, " O'Reilly Media, Inc".
4. Brownley, C. W. (2016). Foundations for Analytics with Python: From Non-Programmer to Hacker. " O'Reilly Media, Inc.".

REFERENCE BOOK

1. Mueller, J. P., & Massaron, L. (2019). Python for data science for dummies. John Wiley & Sons.
2. Grus, J. (2019). Data science from scratch: first principles with python. O'Reilly Media.
3. Kane, F. (2017). Hands-on data science and python machine learning. Packt Publishing Ltd.
4. Madhavan, S. (2015). Mastering python for data science. Packt Publishing Ltd.